

# PAPER\_1342\_QUANTUM\_THERMODYNAMICS

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## PAPER\_1342 — Quantum Thermodynamics Extension

**Framework:** UQFF — Star-Magic v5.27+ | **Tier:** E Quantum Extremes (CLOSED)

**Calculator:** `calculate_paradox({"paradox": "quantum_thermodynamics"})`

**Master Expression** `` Crooks, Jarzynski, Landauer preserved via  $F_U = 1$  `` **\*\*Match: STRUCTURAL\*\***

**Interpretation** All quantum thermo theorems preserved in closed ledger.

**NOT REPLACEMENT** — UQFF integer-primitive identity. Zero fit parameters.

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